Ingenic®

X2600 AIoT Application Processor

Data Sheet

Release 2.1

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北京君正集成电路股份有限公司
Ingenic Semiconductor Co.,Ltd.

Ingenic®

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Data Sheet

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Release history

Date	Revision	Change
Nov.2023	2.1	Modify X2600 0.9V and GND connect on Substrate
Aug.2023	2.0	Modify RISC-V I-Cache and D-Cache size Add DDR3L VDDMEM
May.2023	1.0	First release

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1. Overview

X2600 is a low power consumption, high performance and high integrated application processor featuring a Xburst® 2 Dual-Core and a RISC-V Core. Based on Xburst® 2 dual-core, it integrates H264 decoder, JPEG Codec, 2D graphics engine, RGB/MIPI-DSI/LVDS display, security system and other usefull peripherals.

For more quickly and easily to use X2600, DDR KGD is integrated on chip.

NAME	SIP DDR	VDDMEM(V)
X2600-M	64MB DDR2	1.8
X2600-N	128MB DDR3(L)	1.5/1.35
X2600-E	256MB DDR3(L)	1.5/1.35
X2600-H	512MB DDR3(L)	1.5/1.35

1.1 Block Diagram

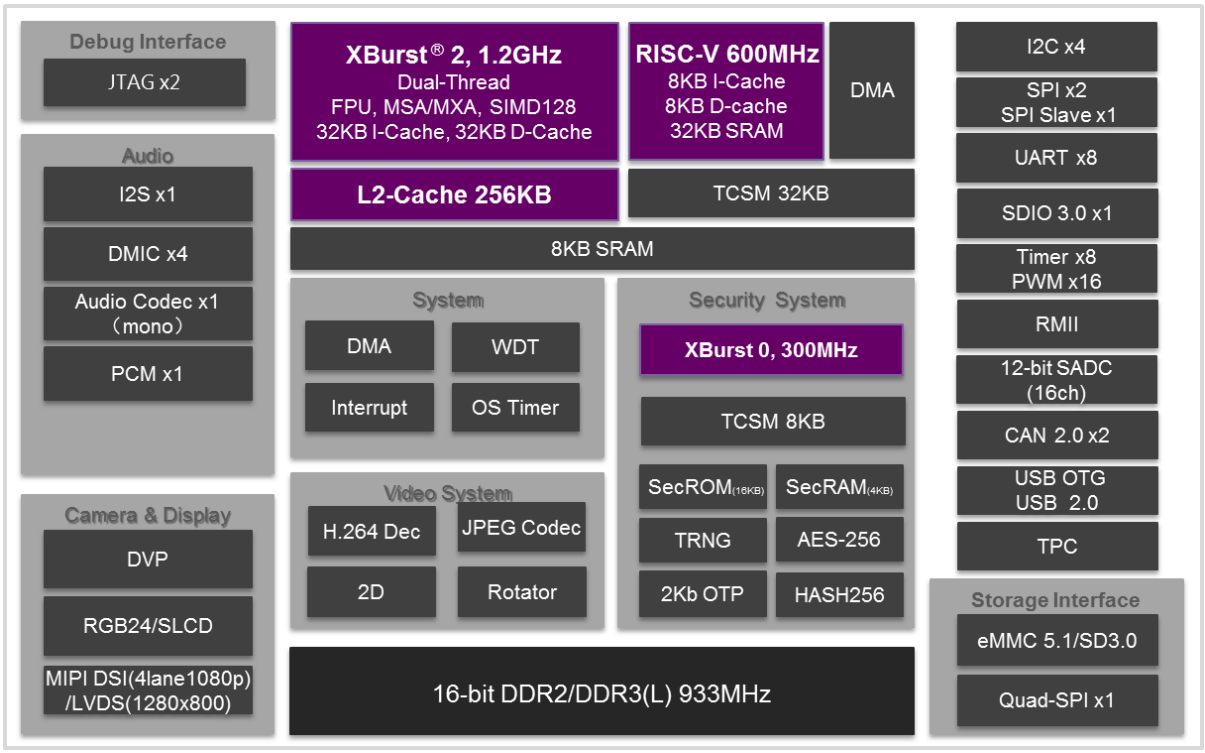


Figure 1-1 X2600 Diagram

1.2 Features

1.2.1 CPU Core

- XBurst® 2, Signal Core, Dual-thread, 1.2Ghz
  - 32KB L1 I-Cache, 32KB L1 D-Cache, 256 KB L2-Cache, 8KB SRAM
  - High-performance Floating-point Unit and SIMD Engine: FSE
    - 32 x 128-bit register set, 128-bit loads/stores to/from SIMD unit
    - IEEE-754 2008 compliant

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- MIPS32 ISA R2 plus MIPS SIMD instruction set architecture:128bit MSA
- Ingenic SIMD instruction set architecture: 128bit MXA
- Support JTAG

- RISC-V(LEP) 600Mhz
  - 8KB L1 D-cache +8K L1 I-cache, 32KB SRAM
  - Support JTAG

1.2.2 Memory Interface

- DDR2/DDR3(L) inside

1.2.3 Video process

- H.264 Decoder
  - Maximum resolution: 2560x2048
  - Decoding capabilities: 1920x1080@60fps
  - Output data format: NV12
- JPEG Decoder
  - Baseline DCT decoder according to JPEG ITU-T T.81 | ISO/IEC 10918-1 standard
  - Unlimited image size (specification up to 64Kx64K)
  - Decoding capabilities: 1920x1080@160fps(ARGB)
  - Input data format：ARGB, NV12, NV21, YUV422, RGB888, YUV444
- JPEG Encoder
  - Baseline DCT compression according to JPEG ITU-T T.81 | ISO/IEC 10918-1.
  - Unlimited image size (specification up to 64Kx64K)
  - Encoding capabilities：1920x1080@110fps(4:2:0 subsampling)
  - Output data format：ARGB, NV12, NV21, YUV422, RGB888, YUV444
- 2D Graphics Engine
  - LineDraw
  - Rectangle Fill
  - BitBLt
  - Color Space Conversion
  - Clipping
  - Alpha Blending
  - Rotation
  - Mirror
  - Scale

1.2.4 Display & Camera

- Display Controller
  - input：ARGB8888, RGB565, RGB555, ARGB1555, NV12, NV21, YUV422
  - output
    - MIPI DSI interface
      - Support 2-lane, 4-lane
      - 1.5Gbps maximum data transfer rate per lane

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        - Capabilities: 1920x1080@60Hz, 24BPP
      - LVDS interface
        - Single channel 8bit
        - Display size up to 1280x800@60Hz, 24BPP
      - RGB interface
        - Support 24bit/pixel(R:8bit, G:8bit, B:8bit)
        - Support 18bit/pixel(R:6bit, G:6bit, B:6bit)
        - Support 16bit/pixel(R:5bit, G:6bit, B:5bit)
        - Capabilities: 1280x720@60Hz, 24BPP
      - 60/80-Type MCU Interface
        - Support 6800, 8080
        - Support 8/9/16/18/24-bit bus
        - Capabilities: 640x480@60Hz, 24BPP
      - 3/4 wire SPI interface
        - Support Serial data transfer interface, 3/4line-spi
        - Support 8/12/16/24-bit bus
  - Maximum resolution: 2047x2047
- Camera Controller
  - Support DVP 8bit input
  - Input data format：RGB888, RGB565, YCbCr 4:2:2, ITU656, RAW
  - Maximum resolution: 2047x2047
  - Capabilities: 640x480@60fps

1.2.5 Audio

- AIC Controller
  - Sample bit support: 24, 20, 18, 16 and 8 bit
  - Sample rate support: 8, 12, 16, 32, 44.1, 48, 96 Khz
  - Support I2S and MSB-Justified format
  - Support Master and Slave mode
- PCM Controller
  - Sample bit support: 16 and 8 bit
  - Sample rate support: 8, 12, 16, 32, 44.1, 48, 96 Khz
  - Support Master and Slave mode
- Internal Audio Codec
  - 24bit DAC with 90db SNR
  - 24bit ADC with 90db SNR
  - Sample bit support: 16, 20 and 24 bit
  - Sample rate support: 8, 12, 16, 32, 44.1, 48, 96 Khz
  - Automatic Level Control (ALC) for smooth audio recording
  - Mono Differential input/output
  - Support loopback
- DMIC Controller
  - Support 1/2/3/4 channel digital MIC

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- SNR: 90dB, THD: -90dB @ FS-20dB
- Sample bit support: 16 and 24 bit
- Sample rate support: 8, 12, 16, 48, 96 Khz
- Support DMIC clock frequency 2.4MHz and 3.072MHz
- Support enable DMIC and AIC at the same time

1.2.6 System Functions

- WatchDog0/1
  - WDT0 generate full reset signal
  - WDT1 generate full interrupt
  - Each WDT has a half interrupt
  - Only EXCLK/512 can be used as the clock for counter
- OS timer
  - One 64bit global timer for system time
  - One 32bit event timer for one logical core
  - Only EXCLK can be used as the clock for counter
- Power On Reset(POR)
  - Monitor 1.8V supply for PLL and 0.9V for core

1.2.7 Security System

- XBurst® 0 MIPS-Based, 300Mhz
- 16KB Secret ROM
- 4KB Secret RAM
- True Random Number Generator
- MD5, SHA, SHA2
- AES 256/192/128-bit key size
- 2Kb OTP(EFUSE)
- Support secure boot

1.2.8 Storage Interface

- High Speed Synchronous serial interfaces (SFC)
  - SPI protocol support: Standard, Dual, Quad SPI
  - 7 transfer formats：Standard SPI, Dual-Input/Dual-Output SPI, Dual-I/O SPI, Full Dual I/O SPI, Quad-Input/Quad-Output SPI, Quad I/O SPI, Full Quad I/O SPI
  - Maximum IO clock frequency of 80MHz
- SD/MMC/SDIO interface(MSC0/1)
  - Support SD 3.0/eMMC 5.1/SDIO 3.0
  - MSC0 support 1/4/8bit，MSC1 support 1/4bit
  - Maximum IO clock frequency of 180Mhz

1.2.9 Peripherals

- General-Purpose I/O ports
  - Each port can be configured as an input, an output or an alternate function port

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- Each port can be configured as an interrupt source triggered by high/low level or rising / falling edge or dual edge
- Support shadow

- I2C x4
  - Three speeds：
    - Standard Mode(100Kb/s)
    - Fast Mode(400Kb/s)
    - High Mode(3.4Mb/s)
  - Master or slave I2C operation
  - 7/10-bit addressing
- SPI x2，SPI Slave x1
  - 3 protocols support: National's Microwire, TI's SSP, and Motorola's SPI
  - Full-duplex or transmit-only or receive-only operation
  - Programmable transfer order: MSB first or LSB first
- UART x8
  - 5-, 6-, 7- or 8-bit characters
  - no parity or even or odd parity
  - 1, 1½, or 2 stop bits
  - Supports modem flow control by software or hardware
  - Slow infrared asynchronous interface that conforms to IrDA1.1 specification
  - Maximum 3M baud rate
  - Support DMA mode
- TCU x2
  - 8 channels each channel has two pins
  - Support posedge / negedge / dualedge clock counting
  - Support gate counting(only count for gating signal)
  - Support direction counting(add / sub because of input signal)
  - Support quadrature counting
  - Support capture counting, output signal high-level time and total cycle time
  - Support exclk / gpio two clock source
- PWM x1
  - 16 channels, output signal ~50MHz, signal precision ~500MHz
  - Cpu / dma mode to update config
- TPC x1
  - One print head control unit, SFT_CLKx1, SFT_DATAx8, LAT_x1, DSTx8
  - Support one 8-channel stepper motor control unit, or two 4-channel stepper motor control unit
  - Support one watchdog timer(WDT) unit
  - output signal ~50MHz, signal precision ~500MHz
- MAC x1
  - 10/100 Mbps operation
  - Supports RMII PHY interfaces
- SAR A/D Converter x1

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- Contains 16 input channels and outputs 12-bit conversion data. Up to 2MS/s sampling rate.
- DNL < 1.5LSB, INL < 3LSB
- Support three conversion sequences.
- Support two conversion modes：
  - Scan conversion mode: Convert multiple analog channels one by one, and the delay time can be configured.
  - Continuous conversion mode:The sequence restarts automatically after all conversions are completed, and the restart interval can be configured.
- Support three trigger modes: Register trigger, TCU trigger and GPIO trigger.
- Setting the priority of the conversion sequences according to requirements can adjust the preemption strategy.
- Support analog watchdog, interrupt is set if the analog voltage converted is below a low threshold or above a high threshold.
- Support DMA transfers.

- CAN2.0 x2
  - Support CAN 2.0B protocol in ISO-11898-1:2003
  - Arbitration Bit Rate up to 1Mbps
  - Support DMA transfers
- USB2.0 x2
  - USB0 support OTG
  - USB1 do not support OTG，but support Device/Host by configuration
  - Fully compliant with USB specifications Rev2.0
    - high-spped（480Mbps）
    - full-speed（12Mbps）
    - low-speed （1.5Mbps）host only
  - port0 support OTG
  - soft connect/disconnect
  - 16 Endpoints
  - Supports control, interrupt, ISO and bulk transfer
  - Support wakeup, remote-wakeup

1.2.10 Bootrom

- 24KB Boot ROM

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2 PAD Information

2.1 Pin Map

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X2600 Ball Assignment Ver1.0

BGA209, 11mm X 11mm X 1.2mm, 0.65mm pitch, top view

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	
A	SFC0_CE__MSC1_D3_PD11	SFC0_DQ2_WP__MSC1_D2_PD10	SFC0_CLK_MSC1_CLK_PD06		TCK_I2C1_SCK_UART1_TXD_PCM_CLK_PC02	DMIC_CLK_UART7_TXD_UART1_RTS__PC04		EFUSE_AVDD	SSI0_CE0__UART1_TXD_RISCV_TMS_I2C1_SCK_PB00		LCD_D22_UART5_RXD_TCU3_IN0_DST2AD14_PB26	LCD_D20_UART4_RXD_TCU2_IN0_DST0AD12_PB24	LCD_D18_UART3_RXD_TCU1_IN0_SFT_D7AD10_PB22		LCD_D13_SLCD_D13_PWM5_SFT_D2_AD5PB17	LCD_D11_SLCD_D11_PWM3_SFT_D0_AD3_PB15	LCD_D9_SLCD_D9_PWM1_DST7_AD1_PB13	A
B	UART2_TXD_PD13	UART2_RXD_PWM_TCU_ADC_TRIG_PD12	SFC0_DQ0_MSC1_CMD_PD07	TDO_I2C0_SCK_UART6_TXD_PCM_DO_PC00	TMS_I2C1_SDA_UART1_RXD_PCM_SYNC_PC03	DMIC_IN0_UART7_RXD_UART1_CTS PC05	RZQ	DDR_PLLVCCA	SSI0_DR_UART1_RXD_RISCV_TDI_I2C1_SDA_PB01	SSI0_CLK_UART1_CTS__RISCV_TCK_I2C2_SCK_PB03	LCD_D21_UART4_TXD_TCU2_IN1_DST1AD13_PB25	LCD_D19_UART3_TXD_TCU1_IN1_LAT__AD11_PB23	LCD_D17_TCU0_IN1_SFT_D6_AD9_PB21	LCD_D15_SLCD_D15_PWM7_SFT_D4_AD7_PB19	LCD_D12_SLCD_D12_PWM4_SFT_D1_AD4PB16	LCD_D8_SLCD_D8_PWM0_AD0_PB12	LCD_D7_SLCD_D7_SM7_PB11	B
C	MSC0_CLK_SFC0_CLK_SSI0_CLK_PD00	MSC0_D1_SFC0_DQ3_HOLD__I2C0_SDAPD03	SFC0_DQ1_MSC1_D0_PD08	SFC0_DQ3_HOLD__MSC1_D1_PD09	TDI_I2C0_SDA_UART6_RXD_PCM_DI_PC01	TRST_	DMIC_IN1_PC06		DDR_ZQ	SSI0_DT_UART1_RTS__RISCV_TDO_I2C2_SDA_PB02	LCD_D23_UART5_TXD_TCU3_IN1_DST3AD15_PB27		LCD_D16_TCU0_IN0_SFT_D5_AD8_PB20	LCD_D14_SLCD_D14_PWM6_SFT_D3_AD6_PB18	LCD_D10_SLCD_D10_PWM2_SFT_CLK_AD2_PB14	LCD_D6_SLCD_D6_SM6_PB10	LCD_D5_SLCD_D5_SM5_PB09	C
D		MSC0_D2_SFC0_DQ2_WP__I2C0_SCK_PD04	MSC0_CMD_SFC0_DQ0_SSI0_DT_PD01												LCD_D4_SLCD_D4_SM4_PB08	LCD_D3_SLCD_D3_SM3_PB07		D
E	MSC0_D4_DRV_VBUS_PE01	MSC0_D3_SFC0_CE__SSI0CE0__PD05	MSC0_D0_SFC0_DQ1_SSI0DR_PD02		VSS	VSS	DDR_VREF	VDDMEM	VDDMEM	VDDMEM	VDDMEM	VSS	VSS		LCD_D2_SLCD_D2_SM2_PB06	LCD_D1_SLCD_D1_SM1_PB05	LCD_D0_SLCD_D0_SM0_PB04	E
F	MSC0_D6_UART7_TXD_PE03	MSC0_D5_EXCLK_PWM10_PE02			VSS	VSS		VDDMEM	VDDMEM	VDDMEM	VDDMEM	VSS	VSS		LCD_HSYNC_SLCD_WR_SSI_SLV_DR_DST6_AVD__PB30	LCD_PCLK_SLCD_CE__SSI_SLV_CLK_DST4_RD__PB28	LCD_VSYNC_SLCD_DC_SSI_SLV_DT_DST5_WE__PB29	F
G	PE00	MSC0_D7_UART7_RXD_PE04	EXCLK_XIN		VSS	VSS	VDD	VSS	VSS	VSS	VSS	VDD			HPOUTPL	HPOUTNL	LCD_DE_SLCD_TE_SSI_SLV_CE__DST7_NEMC_CS1__PB31	G
H		EXCLK_XOUT	AUX7_UART0_CTS__PE12				VDD	VDD		VDD		VDD	VDD			VCM		H
J	AUX5_UART0_RXD_PE10	AUX4_UART0_TXD_PE09	AUX6_UART0_RTS__PE11		VDDIO_SD	VSS		VSS		VSS		VSS	VSS		MICBIAS	MICNL	MICPL	J
K		AUX3_PE08	AUX2_PE07		PLL_AVDD	PLL_AVSS		VDD	VDD	VDD		VSS	CODEC_AVDD		USB1PN	USB1PP		K
L	AUX0_I2C3_SCK_PE05	AUX1_I2C3_SDA_PE06	AUX8_PE13		VDDIO18	VSS						USB_AVD18	VDD		VBUS	USB0PN	USB0PP	L
M	AUX9_PE14	AUX12_PE17			SARADC_AVDD	VSS	VDD	VSS	VSS	VSS	VDDIO33	VSS	USB_AVD33			CIM_HSYNC_PA08	USB0ID	M
N	AUX13_PE18	AUX11_PE16	AUX10_PE15		SARADC_VREFP	VDD	TX_VCCA18	TEST_TE	VSS	VSS	VDDIO33	VSS	VDDIO_CIM		CIM_VSYNC_PA09	CIM_MCLK_PA11	CIM_PCLK_PA10	N
P		AUX15_PE20	AUX14_PE19												CIM_D6_PA06	CIM_D7_PA07		P
R	PPRST_	POR_RST_	BOOT_SEL2_PD16	TX_CLKN	TX_DATAN1		MAC_MDC_UART0_RXD_PC23	MAC_RXD1_PWM_TCU_ADC_TRIG_I2C2_SDA_PC20	MAC_TXD1_I2S0_DO_SSI1_CE0__UART4_TXDPC18	MAC_REF_CLK_I2S0_MCLK_SSI1_DT_UART3_TXD_PC16	PWM9_TCU4_IN1_CAN0_RX_UART5_RXDPC08		MSC1_CLK_SSI1_CLK_PC25	MSC1_D0_SSI1_DR_PC27	CIM_D2_PA02	CIM_D4_PA04	CIM_D5_PA05	R
T	BOOT_SEL1_PD15	TX_DATAN3	TX_DATAN2	TX_CLKP	TX_DATAP1	TX_DATAP0	MAC_RXD0_UART0_CTS__I2C3_SCK_PC21	MAC_TXD0_I2S0_DI_I2C2_SCK_PC19	MAC_PHY_CLK_I2S0_BCLK_SSI1_CLK_UART3_RXD_PC15	PWM8_TCU4_IN0_CAN0_TX_UART5_TXD_PC07	PWM11_TCU5_IN1_CAN1_RX_PC10	PWM13_TCU6_IN1_PC12	PWM15_TCU7_IN1_PC14	MSC1_D1_PC28	MSC1_CMD_SSI1_DT_PC26	CIM_D0_PA00	CIM_D3_PA03	T
U	BOOT_SEL0_PD14	TX_DATAP3	TX_DATAP2		TX_DATAN0	MAC_MDIO_UART0_TXD_PC24	MAC_CRS_DV_UART0_RTS__I2C3_SDA_PC22		MAC_TX_EN_I2S0_LRCK_SSI1_DR_UART4_RXD_PC17		PWM10_TCU5_IN0_CAN1_TX_PC09	PWM12_TCU6_IN0_PC11	PWM14_TCU7_IN0_PC13		MSC1_D3_SSI1_CE0__PC30	MSC1_D2_PC29	CIM_D1_PA01	U
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	

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2.2 Pin Description

2.2.1 GPIO Group A

Ball No.	Ball Name	DS	Pull	Slew Rate	Schmitt	GPIO	Func0	Func1	Func2	Func3	Power
T16	CIM_D0_PA00	8mA	Hiz	fast	No	GPA [0]	CIM_D0				VDDIO_CIM
U17	CIM_D1_PA01	8mA	Hiz	fast	No	GPA [1]	CIM_D1				VDDIO_CIM
R15	CIM_D2_PA02	8mA	Hiz	fast	No	GPA [2]	CIM_D2				VDDIO_CIM
T17	CIM_D3_PA03	8mA	Hiz	fast	No	GPA [3]	CIM_D3				VDDIO_CIM
R16	CIM_D4_PA04	8mA	Hiz	fast	No	GPA [4]	CIM_D4				VDDIO_CIM
R17	CIM_D5_PA05	8mA	Hiz	fast	No	GPA [5]	CIM_D5				VDDIO_CIM
P15	CIM_D6_PA06	8mA	Hiz	fast	No	GPA [6]	CIM_D6				VDDIO_CIM
P16	CIM_D7_PA07	8mA	Hiz	fast	No	GPA [7]	CIM_D7				VDDIO_CIM
M16	CIM_HSYNC_PA08	8mA	Hiz	fast	No	GPA [8]	CIM_HSYNC				VDDIO_CIM
N15	CIM_VSYNC_PA09	8mA	Hiz	fast	No	GPA [9]	CIM_VSYNC				VDDIO_CIM
N17	CIM_PCLK_PA10	8mA	Hiz	fast	No	GPA [10]	CIM_PCLK				VDDIO_CIM
N16	CIM_MCLK_PA11	8mA	Hiz	fast	No	GPA [11]	CIM_MCLK				VDDIO_CIM

2.2.2 GPIO Group B

Ball No.	Ball Name	DS	Pull	Slew Rate	Schmitt	GPIO	Func0	Func1	Func2	Func3	Power
A9	SSI0_CE0__UART1_TXD_RISCV_TMS_I2C1_SCK_PB00	8mA	Hiz	fast	No	GPB [0]	SSI0_CE0_	UART1_TXD	RISCV_TMS	I2C1_SCK	VDDIO33
B9	SSI0_DR_UART1_RXD_RISCV_TDI_I2C1_SDA_PB01	8mA	Hiz	fast	No	GPB [1]	SSI0_DR	UART1_RXD	RISCV_TDI	I2C1_SDA	VDDIO33
C10	SSI0_DT_UART1_RTS__RISCV_TDO_I2C2_SDA_PB02	8mA	Hiz	fast	No	GPB [2]	SSI0_DT	UART1_RTS_	RISCV_TDO	I2C2_SDA	VDDIO33

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B10	SSI0_CLK_UART1_CTS__RISCV_TCK_I2C2_SCK_PB03	8mA	Hiz	fast	No	GPB [3]	SSI0_CLK	UART1_CTS_	RISCV_TCK	I2C2_SCK	VDDIO33
E17	LCD_D0_SLCD_D0_SM0_PB04	8mA	Hiz	fast	No	GPB [4]	LCD_D0	SLCD_D0		SM0	VDDIO33
E16	LCD_D1_SLCD_D1_SM1_PB05	8mA	Hiz	fast	No	GPB [5]	LCD_D1	SLCD_D1		SM1	VDDIO33
E15	LCD_D2_SLCD_D2_SM2_PB06	8mA	Hiz	fast	No	GPB [6]	LCD_D2	SLCD_D2		SM2	VDDIO33
D16	LCD_D3_SLCD_D3_SM3_PB07	8mA	Hiz	fast	No	GPB [7]	LCD_D3	SLCD_D3		SM3	VDDIO33
D15	LCD_D4_SLCD_D4_SM4_PB08	8mA	Hiz	fast	No	GPB [8]	LCD_D4	SLCD_D4		SM4	VDDIO33
C17	LCD_D5_SLCD_D5_SM5_PB09	8mA	Hiz	fast	No	GPB [9]	LCD_D5	SLCD_D5		SM5	VDDIO33
C16	LCD_D6_SLCD_D6_SM6_PB10	8mA	Hiz	fast	No	GPB [10]	LCD_D6	SLCD_D6		SM6	VDDIO33
B17	LCD_D7_SLCD_D7_SM7_PB11	8mA	Hiz	fast	No	GPB [11]	LCD_D7	SLCD_D7		SM7	VDDIO33
B16	LCD_D8_SLCD_D8_PWM0_AD0_PB12	8mA	Hiz	fast	No	GPB [12]	LCD_D8	SLCD_D8	PWM0	AD0	VDDIO33
A17	LCD_D9_SLCD_D9_PWM1_DST7_AD1_PB13	8mA	Hiz	fast	No	GPB [13]	LCD_D9	SLCD_D9	PWM1	DST7/AD1	VDDIO33
C15	LCD_D10_SLCD_D10_PWM2_SFT_CLK_AD2_PB14	8mA	Hiz	fast	No	GPB [14]	LCD_D10	SLCD_D10	PWM2	SFT_CLK/AD2	VDDIO33
A16	LCD_D11_SLCD_D11_PWM3_SFT_D0_AD3_PB15	8mA	Hiz	fast	No	GPB [15]	LCD_D11	SLCD_D11	PWM3	SFT_D0/AD3	VDDIO33
B15	LCD_D12_SLCD_D12_PWM4_SFT_D1_AD4_PB16	8mA	Hiz	fast	No	GPB [16]	LCD_D12	SLCD_D12	PWM4	SFT_D1/AD4	VDDIO33
A15	LCD_D13_SLCD_D13_PWM5_SFT_D2_AD5_PB17	8mA	Hiz	fast	No	GPB [17]	LCD_D13	SLCD_D13	PWM5	SFT_D2/AD5	VDDIO33
C14	LCD_D14_SLCD_D14_PWM6_SFT_D3_AD6_PB18	8mA	Hiz	fast	No	GPB [18]	LCD_D14	SLCD_D14	PWM6	SFT_D3/AD6	VDDIO33
B14	LCD_D15_SLCD_D15_PWM7_SFT_D4_AD7_PB19	8mA	Hiz	fast	No	GPB [19]	LCD_D15	SLCD_D15	PWM7	SFT_D4/AD7	VDDIO33
C13	LCD_D16_TCU0_IN0_SFT_D5_AD8_PB20	8mA	Hiz	fast	No	GPB [20]	LCD_D16	SLCD_RD	TCU0_IN0	SFT_D5/AD8	VDDIO33
B13	LCD_D17_TCU0_IN1_SFT_D6_AD9_PB21	8mA	Hiz	fast	No	GPB [21]	LCD_D17		TCU0_IN1	SFT_D6/AD9	VDDIO33
A13	LCD_D18_UART3_RXD_TCU1_IN0_SFT_D7_AD10_PB22	8mA	Hiz	fast	No	GPB [22]	LCD_D18	UART3_RXD	TCU1_IN0	SFT_D7/AD10	VDDIO33
B12	LCD_D19_UART3_TXD_TCU1_IN1_LAT__AD11_PB23	8mA	Hiz	fast	No	GPB [23]	LCD_D19	UART3_TXD	TCU1_IN1	LAT_/AD11	VDDIO33

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A12	LCD_D20_UART4_RXD_TCU2_IN0_DST0_AD12_PB24	8mA	Hiz	fast	No	GPB [24]	LCD_D20	UART4_RXD	TCU2_IN0	DST0/AD12	VDDIO33
B11	LCD_D21_UART4_TXD_TCU2_IN1_DST1_AD13_PB25	8mA	Hiz	fast	No	GPB [25]	LCD_D21	UART4_TXD	TCU2_IN1	DST1/AD13	VDDIO33
A11	LCD_D22_UART5_RXD_TCU3_IN0_DST2_AD14_PB26	8mA	Hiz	fast	No	GPB [26]	LCD_D22	UART5_RXD	TCU3_IN0	DST2/AD14	VDDIO33
C11	LCD_D23_UART5_TXD_TCU3_IN1_DST3_AD15_PB27	8mA	Hiz	fast	No	GPB [27]	LCD_D23	UART5_TXD	TCU3_IN1	DST3/AD15	VDDIO33
F16	LCD_PCLK_SLCD_CE__SSI_SLV_CLK_DST4_RD__PB28	8mA	Hiz	fast	No	GPB [28]	LCD_PCLK	SLCD_CE_	SSI_SLV_CLK	DST4/RD_	VDDIO33
F17	LCD_VSYNC_SLCD_DC_SSI_SLV_DT_DST5_WE__PB29	8mA	Hiz	fast	No	GPB [29]	LCD_VSYNC	SLCD_DC	SSI_SLV_DT	DST5/WE_	VDDIO33
F15	LCD_HSYNC_SLCD_WR_SSI_SLV_DR_DST6_AVD__PB30	8mA	Hiz	fast	No	GPB [30]	LCD_HSYNC	SLCD_WR	SSI_SLV_DR	DST6/AVD_	VDDIO33
G17	LCD_DE_SLCD_TE_SSI_SLV_CE__DST7_NEMC_CS1__PB31	8mA	Hiz	fast	No	GPB [31]	LCD_DE	SLCD_TE	SSI_SLV_CE_	DST7/NEMC_CS1_	VDDIO33

2.2.3 GPIO Group C

Ball No.	Ball Name	DS	Pull	Slew Rate	Schmitt	GPIO	Func0	Func1	Func2	Func3	Power
B4	TDO_I2C0_SCK_UART6_TXD_PCM_DO_PC00	8mA	Hiz	fast	No	GPC [0]	TDO	I2C0_SCK	UART6_TXD	PCM_DO	VDDIO33
C5	TDI_I2C0_SDA_UART6_RXD_PCM_DI_PC01	8mA	PU	fast	No	GPC [1]	TDI	I2C0_SDA	UART6_RXD	PCM_DI	VDDIO33
A5	TCK_I2C1_SCK_UART1_TXD_PCM_CLK_PC02	8mA	PU	fast	No	GPC [2]	TCK	I2C1_SCK	UART1_TXD	PCM_CLK	VDDIO33
B5	TMS_I2C1_SDA_UART1_RXD_PCM_SYNC_PC03	8mA	PU	fast	No	GPC [3]	TMS	I2C1_SDA	UART1_RXD	PCM_SYNC	VDDIO33
A6	DMIC_CLK_UART7_TXD_UART1_RTS__PC04	8mA	Hiz	fast	No	GPC [4]	DMIC_CLK	UART7_TXD	UART1_RTS_		VDDIO33
B6	DMIC_IN0_UART7_RXD_UART1_CTS__PC05	8mA	Hiz	fast	No	GPC [5]	DMIC_IN0	UART7_RXD	UART1_CTS_		VDDIO33
C7	DMIC_IN1_PC06	8mA	Hiz	fast	No	GPC [6]	DMIC_IN1				VDDIO33

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T10	PWM8_TCU4_IN0_CAN0_TX_UART5_TXD_PC07	8mA	Hiz	fast	No	GPC [7]	PWM8	TCU4_IN0	CAN0_TX	UART5_TXD	VDDIO33
R11	PWM9_TCU4_IN1_CAN0_RX_UART5_RXD_PC08	8mA	Hiz	fast	No	GPC [8]	PWM9	TCU4_IN1	CAN0_RX	UART5_RXD	VDDIO33
U11	PWM10_TCU5_IN0_CAN1_TX_PC09	8mA	Hiz	fast	No	GPC [9]	PWM10	TCU5_IN0	CAN1_TX		VDDIO33
T11	PWM11_TCU5_IN1_CAN1_RX_PC10	8mA	Hiz	fast	No	GPC [10]	PWM11	TCU5_IN1	CAN1_RX		VDDIO33
U12	PWM12_TCU6_IN0_PC11	8mA	Hiz	fast	No	GPC [11]	PWM12	TCU6_IN0			VDDIO33
T12	PWM13_TCU6_IN1_PC12	8mA	Hiz	fast	No	GPC [12]	PWM13	TCU6_IN1			VDDIO33
U13	PWM14_TCU7_IN0_PC13	8mA	Hiz	fast	No	GPC [13]	PWM14	TCU7_IN0			VDDIO33
T13	PWM15_TCU7_IN1_PC14	8mA	PD	fast	No	GPC [14]	PWM15	TCU7_IN1			VDDIO33
T9	MAC_PHY_CLK_I2S0_BCLK_SSI1_CLK_UART3_RXD_PC15	8mA	Hiz	fast	No	GPC [15]	MAC_PHY_CLK	I2S0_BCLK	SSI1_CLK	UART3_RXD	VDDIO33
R10	MAC_REF_CLK_I2S0_MCLK_SSI1_DT_UART3_TXD_PC16	8mA	Hiz	fast	No	GPC [16]	MAC_REF_CLK	I2S0_MCLK	SSI1_DT	UART3_TXD	VDDIO33
U9	MAC_TX_EN_I2S0_LRCK_SSI1_DR_UART4_RXD_PC17	8mA	Hiz	fast	No	GPC [17]	MAC_TXEN	I2S0_LRCK	SSI1_DR	UART4_RXD	VDDIO33
R9	MAC_TXD1_I2S0_DO_SSI1_CE0__UART4_TXD_PC18	8mA	Hiz	fast	No	GPC [18]	MAC_TXD1	I2S0_DO	SSI1_CE0_	UART4_TXD	VDDIO33
T8	MAC_TXD0_I2S0_DI_I2C2_SCK_PC19	8mA	Hiz	fast	No	GPC [19]	MAC_TXD0	I2S0_DI	I2C2_SCK		VDDIO33
R8	MAC_RXD1_PWM_TCU_ADC_TRIG_I2C2_SDA_PC20	8mA	Hiz	fast	No	GPC [20]	MAC_RXD1	PWM_TCU_ADC_TRIG	I2C2_SDA		VDDIO33
T7	MAC_RXD0_UART0_CTS__I2C3_SCK_PC21	8mA	Hiz	fast	No	GPC [21]	MAC_RXD0	UART0_CTS_	I2C3_SCK		VDDIO33
U7	MAC_CRS_DV_UART0_RTS__I2C3_SDA_PC22	8mA	Hiz	fast	No	GPC [22]	MAC_CRS_DV	UART0_RTS_	I2C3_SDA		VDDIO33
R7	MAC_MDC_UART0_RXD_PC23	8mA	Hiz	fast	No	GPC [23]	MAC_MDC	UART0_RXD			VDDIO33
U6	MAC_MDIO_UART0_TXD_PC24	8mA	Hiz	fast	No	GPC [24]	MAC_MDIO	UART0_TXD			VDDIO33
R13	MSC1_CLK_SSI1_CLK_PC25	8mA	PU	fast	No	GPC [25]	MSC1_CLK	SSI1_CLK			VDDIO33
T15	MSC1_CMD_SSI1_DT_PC26	8mA	PU	fast	No	GPC [26]	MSC1_CMD	SSI1_DT			VDDIO33
R14	MSC1_D0_SSI1_DR_PC27	8mA	PU	fast	No	GPC [27]	MSC1_D0	SSI1_DR			VDDIO33
T14	MSC1_D1_PC28	8mA	PU	fast	No	GPC [28]	MSC1_D1				VDDIO33

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U16	MSC1_D2_PC29	8mA	PU	fast	No	GPC [29]	MSC1_D2				VDDIO33
U15	MSC1_D3_SSI1_CE0__PC30	8mA	PU	fast	No	GPC [30]	MSC1_D3	SSI1_CE0_			VDDIO33

2.2.4 GPIO Group D

Ball No.	Ball Name	DS	Pull	Slew Rate	Schmitt	GPIO	Func0	Func1	Func2	Func3	Power
C1	MSC0_CLK_SFC0_CLK_SSI0_CLK_PD00	26pF	PU	fast	No	GPD [0]	MSC0_CLK	SFC0_CLK	SSI0_CLK		VDDIO_SD
D3	MSC0_CMD_SFC0_DQ0_SSI0_DT_PD01	26pF	PU	fast	No	GPD [1]	MSC0_CMD	SFC0_DQ0	SSI0_DT		VDDIO_SD
E3	MSC0_D0_SFC0_DQ1_SSI0_DR_PD02	26pF	PU	fast	No	GPD [2]	MSC0_D0	SFC0_DQ1	SSI0_DR		VDDIO_SD
C2	MSC0_D1_SFC0_DQ3_HOLD__I2C0_SDA_PD03	26pF	PU	fast	No	GPD [3]	MSC0_D1	SFC0_DQ3/HOLD_	I2C0_SDA		VDDIO_SD
D2	MSC0_D2_SFC0_DQ2_WP__I2C0_SCK_PD04	26pF	PU	fast	No	GPD [4]	MSC0_D2	SFC0_DQ2/WP_	I2C0_SCK		VDDIO_SD
E2	MSC0_D3_SFC0_CE__SSI0_CE0__PD05	26pF	PU	fast	No	GPD [5]	MSC0_D3	SFC0_CE_	SSI0_CE0_		VDDIO_SD
A3	SFC0_CLK_MSC1_CLK_PD06	8mA	PU	fast	No	GPD [6]	SFC0_CLK	MSC1_CLK			VDDIO33
B3	SFC0_DQ0_MSC1_CMD_PD07	8mA	PU	fast	No	GPD [7]	SFC0_DQ0	MSC1_CMD			VDDIO33
C3	SFC0_DQ1_MSC1_D0_PD08	8mA	PU	fast	No	GPD [8]	SFC0_DQ1	MSC1_D0			VDDIO33
C4	SFC0_DQ3_HOLD__MSC1_D1_PD09	8mA	PU	fast	No	GPD [9]	SFC0_DQ3/HOLD_	MSC1_D1			VDDIO33
A2	SFC0_DQ2_WP__MSC1_D2_PD10	8mA	PU	fast	No	GPD [10]	SFC0_DQ2/WP_	MSC1_D2			VDDIO33
A1	SFC0_CE__MSC1_D3_WAIT__PD11	8mA	PU	fast	No	GPD [11]	SFC0_CE_	MSC1_D3		WAIT_	VDDIO33
B2	UART2_RXD_PWM_TCU_ADC_TRIG_PD12	8mA	PU	fast	No	GPD [12]	UART2_RXD	PWM_TCU_ADC_TRIG			VDDIO33
B1	UART2_TXD_PD13	8mA	Hiz	fast	No	GPD [13]	UART2_TXD				VDDIO33

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U1	BOOT_SEL0_PD14	8mA	Hiz	fast	Yes	GPD [14]	BOOT_SEL0				VDDIO33
T1	BOOT_SEL1_PD15	8mA	Hiz	fast	Yes	GPD [15]	BOOT_SEL1				VDDIO33
R3	BOOT_SEL2_PD16	8mA	PD	fast	Yes	GPD [16]	BOOT_SEL2				VDDIO33

2.2.5 GPIO Group E

Ball No.	Ball Name	DS	Pull	Slew Rate	Sch mitt	GPIO	Func0	Func1	Func2	Func3	Power
G1	PE00	12mA	PD	dis	No	GPE [0]					VDDIO18
E1	MSC0_D4_DRV_VBUS_PE01	12mA	Hiz	dis	No	GPE [1]	MSC0_D4	DRV_VBUS			VDDIO18
F2	MSC0_D5_EXCLK_PWM10_PE02	12mA	Hiz	dis	No	GPE [2]	MSC0_D5	EXCLK	PWM10		VDDIO18
F1	MSC0_D6_UART7_TXD_PE03	12mA	Hiz	dis	No	GPE [3]	MSC0_D6	UART7_TXD			VDDIO18
G2	MSC0_D7_UART7_RXD_PE04	12mA	Hiz	dis	No	GPE [4]	MSC0_D7	UART7_RXD			VDDIO18
L1	AUX0_I2C3_SCK_PE05	12mA	Hiz	dis	No	GPE [5]	AUX0	I2C3_SCK			SARADC_AVDD
L2	AUX1_I2C3_SDA_PE06	12mA	Hiz	dis	No	GPE [6]	AUX1	I2C3_SDA			SARADC_AVDD
K3	AUX2_PE07	12mA	Hiz	dis	No	GPE [7]	AUX2				SARADC_AVDD
K2	AUX3_PE08	12mA	Hiz	dis	No	GPE [8]	AUX3				SARADC_AVDD
J2	AUX4_UART0_TXD_PE09	12mA	Hiz	dis	No	GPE [9]	AUX4	UART0_TXD			SARADC_AVDD
J1	AUX5_UART0_RXD_PE10	12mA	Hiz	dis	No	GPE [10]	AUX5	UART0_RXD			SARADC_AVDD
J3	AUX6_UART0_RTS__PE11	12mA	Hiz	dis	No	GPE [11]	AUX6	UART0_RTS			SARADC_AVDD
H3	AUX7_UART0_CTS__PE12	12mA	Hiz	dis	No	GPE [12]	AUX7	UART0_CTS			SARADC_AVDD
L3	AUX8_PE13	12mA	Hiz	dis	No	GPE [13]	AUX8				SARADC_AVDD
M1	AUX9_PE14	12mA	Hiz	dis	No	GPE [14]	AUX9				SARADC_AVDD
N3	AUX10_PE15	12mA	Hiz	dis	No	GPE [15]	AUX10				SARADC_AVDD
N2	AUX11_PE16	12mA	Hiz	dis	No	GPE [16]	AUX11				SARADC_AVDD
M2	AUX12_PE17	12mA	Hiz	dis	No	GPE [17]	AUX12				SARADC_AVDD
N1	AUX13_PE18	12mA	Hiz	dis	No	GPE [18]	AUX13				SARADC_AVDD
P3	AUX14_PE19	12mA	Hiz	dis	No	GPE [19]	AUX14				SARADC_AVDD
P2	AUX15_PE20	12mA	Hiz	dis	No	GPE [20]	AUX15				SARADC_AVDD

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2.3 Analog PAD Description

Ball No.	Pin Names	IO	Power	Pin Description
Memory				
B8	DDR_PLLVCCA	P	-	DDR PHY PLL Supply 1.8V
B7	RZQ	I	VDDMEM	For DDR, external reference resistor for output calibrating
C9	DDR_ZQ	I	VDDMEM	For DDR, external reference resistor for output calibrating
E7	DDR_VREF	P	-	Reference voltage supply of DDR , typical 0.5*VDDMEM
Power and Ground				
E8	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
E9	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
E10	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
E11	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
F8	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
F9	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
F10	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
F11	VDDMEM	P	-	DDR PHY IO Supply, 1.5V for DDR3, 1.35 for DDR3L, 1.8V for DDR2
M11	VDDIO33	P	-	IO digital power for GPIO Port B&C&D, 3.3V
N11	VDDIO33	P	-	IO digital power for GPIO Port B&C&D, 3.3V
L5	VDDIO18	P	-	IO digital power for GPIO Port E, 1.8V
N13	VDDIO_CIM	P	-	CIM GPIO Supply 1.8/3.3V
J5	VDDIO_SD	P	-	SD GPIO Supply 1.8/3.3V
E5	VSS	P	-	Ground
E6	VSS	P	-	Ground
E12	VSS	P	-	Ground
E13	VSS	P	-	Ground
F5	VSS	P	-	Ground
F6	VSS	P	-	Ground
F12	VSS	P	-	Ground

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F13	VSS	P	-	Ground
G5	VSS	P	-	Ground
G6	VSS	P	-	Ground
G8	VSS	P	-	Ground
G9	VSS	P	-	Ground
G10	VSS	P	-	Ground
G11	VSS	P	-	Ground
J6	VSS	P	-	Ground
J8	VSS	P	-	Ground
J10	VSS	P	-	Ground
J12	VSS	P	-	Ground
J13	VSS	P	-	Ground
K12	VSS	P	-	Ground
L6	VSS	P	-	Ground
M6	VSS	P	-	Ground
M8	VSS	P	-	Ground
M9	VSS	P	-	Ground
M10	VSS	P	-	Ground
M12	VSS	P	-	Ground
N9	VSS	P	-	Ground
N10	VSS	P	-	Ground
N12	VSS	P	-	Ground
G7	VDD	P	-	Power supply 0.9V
G12	VDD	P	-	Power supply 0.9V
H7	VDD	P	-	Power supply 0.9V
H8	VDD	P	-	Power supply 0.9V
H10	VDD	P	-	Power supply 0.9V
H12	VDD	P	-	Power supply 0.9V
H13	VDD	P	-	Power supply 0.9V
K8	VDD	P	-	Power supply 0.9V
K9	VDD	P	-	Power supply 0.9V
K10	VDD	P	-	Power supply 0.9V

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L13	VDD	P	-	Power Supply 0.9V
M7	VDD	P	-	Power Supply 0.9V
N6	VDD	P	-	Power Supply 0.9V
DSI-TX				
N7	TX_VCCA18	P	-	MIPI DSI Analog IO Supply 1.8V
U5	TX_DATAN0	AIO	TX_VCCA18	Lane0 negative end
T6	TX_DATAP0	AIO	TX_VCCA18	Lane0 positive end
R5	TX_DATAN1	AO	TX_VCCA18	Lane1 negative end
T5	TX_DATAP1	AO	TX_VCCA18	Lane1 positive end
T3	TX_DATAN2	AO	TX_VCCA18	Lane2 negative end
U3	TX_DATAP2	AO	TX_VCCA18	Lane2 positive end
T2	TX_DATAN3	AO	TX_VCCA18	Lane3 negative end
U2	TX_DATAP3	AO	TX_VCCA18	Lane3 positive end
R4	TX_CLKN	AO	TX_VCCA18	CLK lane negative end
T4	TX_CLKP	AO	TX_VCCA18	CLK lane positive end
Audio Codec				
K13	CODEC_AVDD	P	-	Codec Analog Supply 1.8V
H16	VCM	AO	CODEC_AVDD	Reference voltage output
J17	MICPL	AI	CODEC_AVDD	ADC channel input
J16	MICNL	AI	CODEC_AVDD	ADC channel input
J15	MICBIAS	AO	CODEC_AVDD	Microphone bias output
G16	HPOUTNL	AO	CODEC_AVDD	DAC channel output
G15	HPOUTPL	AO	CODEC_AVDD	DAC channel output
USB OTG				
M13	USB_AVD33	P	-	USB Analog Supply.3.3V
L12	USB_AVD18	P	-	USB Analog Supply.1.8V
L16	USB0PN	AIO	USB_AVD33	USB0 OTG data negative
L17	USB0PP	AIO	USB_AVD33	USB0 OTG data positive
L15	USB0VBUS	AI	USB_AVD33	The USB0VBUS is used for signaling to detect which device is connected. No charge pumps inside PHY, so no supplying capability by default. The PHY

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				supports the VBUS divided to 3/5(default) off chip or not.
M17	USB0ID	AI	USB_AVD18	Used to identify the device attached to the PHY. The state of the pin is one of: high impedance (>1M Ω), or low impedance (<10Ω to ground).
K15	USB1PN	AIO	USB_AVD33	USB1 data negative
K16	USB1PP	AIO	USB_AVD33	USB1 data positive
SARADC				
M5	SARADC_AVDD	P	-	SARADC Analog Supply 1.8V
N5	SARADC_VREFP	AI	SARADC_AVDD	Positive reference voltage input
L1	AUX0	AI	SARADC_AVDD	SARADC input channel 0
L2	AUX1	AI	SARADC_AVDD	SARADC input channel 1
K3	AUX2	AI	SARADC_AVDD	SARADC input channel 2
K2	AUX3	AI	SARADC_AVDD	SARADC input channel 3
J2	AUX4	AI	SARADC_AVDD	SARADC input channel 4
J1	AUX5	AI	SARADC_AVDD	SARADC input channel 5
J3	AUX6	AI	SARADC_AVDD	SARADC input channel 6
H3	AUX7	AI	SARADC_AVDD	SARADC input channel 7
L3	AUX8	AI	SARADC_AVDD	SARADC input channel 8
M1	AUX9	AI	SARADC_AVDD	SARADC input channel 9
N3	AUX10	AI	SARADC_AVDD	SARADC input channel 10
N2	AUX11	AI	SARADC_AVDD	SARADC input channel 11
M2	AUX12	AI	SARADC_AVDD	SARADC input channel 12
N1	AUX13	AI	SARADC_AVDD	SARADC input channel 13
P3	AUX14	AI	SARADC_AVDD	SARADC input channel 14
P2	AUX15	AI	SARADC_AVDD	SARADC input channel 15
EFUSE				
A8	EFUSE_AVDD	P	-	EFUSE programming power, 1.8V
CPM				
G3	EXCLK_XIN	AI	VDDIO18	EXCLK OSC Input
H2	EXCLK_XOUT	AO	VDDIO18	EXCLK OSC Output
K5	PLL_AVDD	P	-	PLL Analog Supply 1.8V
K6	PLL_AVSS	P	-	PLL Analog Ground

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2.4 Digital PAD Description

Ball No.	Pin Names	IO	Power	Pin Description
System Control				
R1	PPRST_	I	VDDIO33	power on reset and RESET-KEY reset input
C6	TRST_	I	VDDIO33	JTAG reset
N8	TEST_TE	I	VDDIO33	Manufacture test enable, program readable
R2	POR_RST_	O	VDDIO33	POR Reset Output(open drain)

Pin Names	IO	Pin Description
LCD		
LCD_PCLK	O	LCD pixel clock
LCD_VSYNC	O	LCD frame sync
LCD_HSYNC	O	LCD line sync
LCD_DE	O	LCD data enable
LCD_D<n>	O	LCD data output bit n (0-23)
SLCD(60/80-Type MCU)		
SLCD_WR	O	Smart LCD write signal
SLCD_CE_	O	Smart LCD chip select signal
SLCD_TE	I	Smart LCD tearing effect signal
SLCD_DC	O	Smart LCD data/command select signal
SLCD_D<n>	O	Smart LCD data output bit n (0-15)
CIM(Camera Interface)		
CIM_PCLK	I	CIM pixel clock input
CIM_HSYNC	I	CIM line horizontal sync input

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CIM_VSYNC	I	CIM vertical sync input
CIM_MCLK	O	CIM master clock output
CIM_D<n>	I	CIM data input bit n (0-7)
I2S		
I2S0_MCLK	O	I2S master clock output
I2S0_BCLK	IO	I2S bit clock
I2S0_LRCK	IO	I2S LR clock
I2S0_DO	O	I2S data output
I2S0_DI	I	I2S data input
DMIC		
DMIC_CLK	O	Digital MIC clock output
DMIC_IN0	I	DMIC data in for DMIC 0/1
DMIC_IN1	I	DMIC data in for DMIC 2/3
PCM		
PCM_CLK	IO	PCM clock
PCM_SYNC	IO	PCM sync
PCM_DO	O	PCM data out
PCM_DI	I	PCM data in
SFC		
SFC0_CLK	O	Serial Flash clock output
SFC0_CE_	O	Serial Flash chip enable
SFC0_DQ0	IO	Serial Flash data
SFC0_DQ1	IO	Serial Flash data
SFC0_DQ2/WP_	IO	Serial Flash write protect signal
SFC0_DQ3/HOLD_	IO	Serial Flash hold signal
MSC		
MSC<n>_CLK	O	MSC(MMC/SD) n clock output
MSC<n>_CMD	IO	MSC(MMC/SD) n command
MSC<n>_D<m>	IO	MSC(MMC/SD) n data bit m (0-7)
PWM		
PWM<n>	O	PWM n data output

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PWM_TCU_TRIG	I	Use to save PWM counter
TCU		
TCU<n>_IN0	I	TCU n data input
TCU<n>_IN1	I	TCU n data input
PWM_TCU_ADC_TRIG	I	Use to save TCU counter
UART		
UART<n>_RXD	I	UART n receiving data
UART<n>_TXD	O	UART n transmitting data
UART<n>_CTS_	I	UART n Clear to send control
UART<n>_RTS_	O	UART n Request to send control
I2C		
I2C<n>_SCK	IO	I2C n serial clock
I2C<n>_SDA	IO	I2C n serial data
SSI		
SSI0_CLK	O	SSI clock output
SSI0_CE0_	O	SSI chip enable 0
SSI0_DT	O	SSI data output
SSI0_DR	I	SSI data input
SSI_Slave		
SSI_SLV_CLK	I	SSI Slave clock
SSI_SLV_CE0_	I	SSI Slave chip enable
SSI_SLV_DT	O	SSI Slave data output
SSI_SLV_DR	I	SSI Slave data input
NEMC(LocalBus)		
NEMC_CS1_	O	NEMC chip select1
AVD_	O	Address Valid ouput
RD_	O	NEMC read enable, low active
WE_	O	NEMC write enable, low active
ADn	IO	Address[15:0], Data[15:0] in muxed mode

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WAIT_	I	External wait state request signal
CAN		
CAN<n>_TX	O	CAN n TX output to CAN transceiver
CAN<n>_RX	I	CAN n RX input from CAN transceiver
TPC		
SFT_CLK	O	Thermal Print Head data clock
SFT_D<n>	O	Thermal Print Head data channel n(0-7)
LAT_	O	Thermal Print Head latch pin
DST<n>	O	Thermal Print Head heating channel n(0-7)
SM<n>	O	Stepper Motor unit channel n(0-7)
DEBUG		
TDO	O	JTAG serial data output
TDI	I	JTAG serial data input
TCK	I	JTAG clock
TMS	I	JTAG mode select
RISCV_TDO	O	JTAG serial data output for RISC-V
RISCV_TDI	I	JTAG serial data input for RISC-V
RISCV_TCK	I	JTAG clock for RISC-V
RISCV_TMS	I	JTAG mode select for RISC-V

NOTES:

1 The meaning of phases in IO cell characteristics are:

a DS: default output drive strength

b Pull: default driver state, PD(pull-up), PU(pull-down) , HIZ

c SlewRate: slew-rate function select fast mode by default

d Schmitt: The IO cell is Schmitt trigger input by default

e I: Digtal Input

f O: Digtal Output

g IO: Digtal Input and Output

h AI: Analog Input

i AO: Analog Output

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- j AIO: Analog Input and Output

2 All GPIO shared pins are reset to GPIO input except PC00-03 is function 0

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3 Electrical Specifications

3.1 Absolute Maximum Ratings

The absolute maximum ratings for the processors are listed in Table 3 1. Do not exceed these parameters or the part may be damaged permanently. Operation at absolute maximum ratings is not guaranteed.

Table 3-1 Absolute Maximum Rating

Parameter	Min	Max	Unit
Storage Temperature	-65	150	°C
Operation Temperature	-40	85	°C
VDD core power supplies voltage	-0.5	1.05	V
VDDMEM power supplies voltage(1.5V for DDR3)	-0.4	1.8	V
VDDMEM power supplies voltage(1.35V for DDR3L)	-0.4	1.8	V
VDDMEM power supplies voltage(1.8V for DDR2)	-0.5	2.3	V
DDR_PLLVCCA power supplies voltage	-0.5	1.98	V
PLL_AVDD power supplies voltage	-0.5	1.98	V
VDDIO18 power supplies voltage	-0.5	1.98	V
VDDIO33 power supplies voltage	-0.5	3.63	V
VDDIO_CIM power supplies voltage	-0.5	3.63	V
VDDIO_SD power supplies voltage	-0.5	3.63	V
TX_VCCA18	-0.1	1.98	V
CODEC_AVDD	-0.1	1.98	V
USB_AVD33 power supplies voltage	-0.1	3.63	V
USB_AVD18 power supplies voltage	-0.1	1.98	V
SARADC_AVDD	-0.1	1.98	V
EFUSE_AVDD power supplies voltage	-0.1	1.98	V
IO voltage to VDDMEM(1.5V for DDR3) supplied non-supply pins	-0.4	1.8	V
IO voltage to VDDMEM(1.35V for DDR3L) supplied non-supply pins	-0.4	1.8	V
IO voltage to VDDMEM(1.8V for DDR2) supplied non-supply pins	-0.5	2.3	V
IO voltage to VDDIO18 supplied non-supply pins	-0.3	2.28	V
IO voltage to VDDIO33 supplied non-supply pins	-0.5	3.93	V
IO voltage to VDDIO_CIM supplied non-supply pins	-0.5	3.93	V
IO voltage to VDDIO_SD supplied non-supply pins	-0.5	3.93	V
IO voltage to CODEC_AVDD supplied non-supply pins	-0.1	1.98	V
IO voltage to USB_AVD33 supplied non-supply pins	-0.1	3.63	V
input voltage to SARADC_AVDD supplied non-supply pins	-0.1	1.98	V
Maximum ESD stress voltage, Human Body Model; Any pin to any supply pin, either polarity, or Any pin to all non-supply pins together, either polarity. Three stresses maximum.	-	2000	V

3.2 Recommended operating conditions

***Table 3-2 Recommended operating con*ditions for power supplies**

Symbol	Description	Min	Typical	Max	Unit
VCORE	VDD core voltage	0.9	0.9	0.99	V
VMEM(1.5V)	VDDMEM voltage for DDR3	1.425	1.5	1.575	V
VMEM(1.35V)	VDDMEM voltage for DDR3L	1.283	1.35	1.45	V
VMEM(1.8V)	VDDMEM voltage for DDR2	1.71	1.8	1.89	V

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VDDRPLL	DDR_PLLVCCA voltage	1.62	1.8	1.98	V
VPLLAVDD	PLL_AVDD voltage	1.62	1.8	1.98	V
VIO18	VDDIO18 voltage	1.62	1.8	1.98	V
VIO33	VDDIO33 voltage	2.97	3.3	3.63	V
VCIM(1.8V)	VDDIO_CIM voltage, use as 1.8V	1.62	1.8	1.98	V
VCIM(3.3V)	VDDIO_CIM voltage, use as 3.3V	2.97	3.3	3.63	V
VSD(1.8V)	VDDIO_SD voltage, use as 1.8V	1.7	1.8	1.95	V
VSD (3.3V)	VDDIO_SD voltage, use as 3.3V	2.7	3.3	3.63	V
VDSI18	TX_VCCA18 voltage	1.62	1.8	1.98	V
VCDC	CODEC_AVDD voltage	1.62	1.8	1.98	V
VUSB33	USB_AVD33 voltage	3.0	3.3	3.6	V
VUSB18	USB_AVD18 voltage	1.62	1.8	1.98	V
VADC	SARADC_AVDD voltage	1.62	1.8	1.98	V
VEFUSE	EFUSE_AVDD voltage	1.71	1.8	1.98	V

Table 3-3 Recommended operating conditions for SARADC pins

Symbol	Description	Min	Typical	Max	Unit
SARADC_VREFP	SARADC_VREFP and SARADC_AGND are used as reference for SARADC	-	1.8	VADC	V
V_{IADC}	AUX0-15 input voltage range	0		VADC	V

Table 3-4 Recommended operating conditions for others

Symbol	Description	Min	Typical	Max	Unit
TA	Ambient temperature	–40		85	°C

3.3 DC Specifications

The DC characteristics for each pin include input-sense levels and output-drive levels and currents. These parameters can be used to determine maximum DC loading, and also to determine maximum transition times for a given load. All DC specification values are valid for the entire temperature range of the device

Table 3-5 DC characteristics for VREF

Symbol	Parameter	Min	Typical	Max	Unit
DDR_VREF(DC)	Reference voltage supply of DDR	0.49* VMEM	0.5* VMEM	0.51* VMEM	V

NOTE: The ac peak noise on VREF may not allow VREF to deviate from VREF(DC) by more than +/- 1% VMEM.

Table 3-6 DC characteristics for VDDIO33/VDDIO_CIM supplied pins for 3.3V application

Symbol	Parameter	Min	Typical	Max	Unit
V_{DVDD}	I/O supply voltage	2.97	3.3	3.63	V
V_{PAD}	Voltage at IO	-0.3		V_{DVDD}+0.3	V
V_{IL}	Input Low Voltage	-0.3		0.8	V
V_{IH}	Input High Voltage	2		V_{DVDD}+0.3	V
V_{T+}	Schmitt trig low to high threshold point	0.9		2.1	V

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V_{T-}	Schmitt trig high to low threshold point		0.7		1.9	V
R_{PU}	Pull-up Resistor		36	57	100	kΩ
R_{PD}	Pull-down Resistor		35	57	103	kΩ
V_{OL}	Output low voltage				0.23	V
V_{OH}	Output high voltage		2.69			V
I_{OL}	Low level output current	2mA	3.5	6.0	8.8	mA
		4mA	8.1	13.0	20.5	mA
		8mA	13.8	23.8	35.2	mA
		12mA	18.4	31.7	46.9	mA
I_{OH}	High level output current	2mA	3.4	5.3	7.5	mA
		4mA	6.9	10.7	14.9	mA
		8mA	12.7	19.7	27.5	mA
		12mA	16.1	25.0	35.0	mA

Table 3-7 DC characteristics for VDDIO_CIM supplied pins for 1.8V application

Symbol	Parameter		Min	Typical	Max	Unit
V_{DVDD}	I/O supply voltage		1.62	1.8	1.98	V
V_{PAD}	Voltage at IO		-0.3		V_{DVDD}+0.3	V
V_{IL}	Input Low Voltage		-0.3		0.35*V_{DVDD}	V
V_{IH}	Input High Voltage		0.65*V_{DVDD}		V_{DVDD}+0.3	V
V_{T+}	Schmitt trig low to high threshold point		0.4*V_{DVDD}		0.7*V_{DVDD}	V
V_{T-}	Schmitt trig high to low threshold point		0.3*V_{DVDD}		0.6*V_{DVDD}	V
R_{PU}	Pull-up Resistor		35	54	88	kΩ
R_{PD}	Pull-down Resistor		35	59	114	kΩ
V_{OL}	Output low voltage				0.24	V
V_{OH}	Output high voltage		1.35			V
I_{OL}	Low level output current	2mA	3.4	6.2	9.2	mA
		4mA	7.9	14.4	21.4	mA
		8mA	13.6	24.7	36.8	mA
		12mA	18.1	32.9	49.0	mA
I_{OH}	High level output current	2mA	3.6	5.5	7.6	mA
		4mA	7.3	11.1	15.1	mA
		8mA	13.4	20.4	27.9	mA
		12mA	17.0	26.0	35.5	mA

Table 3-8 DC characteristics for VDDIO_SD supplied pins for 3.3V application

Symbol	Parameter	Min	Typical	Max	Unit
V_{DVDD}	I/O supply voltage	2.7	3.3	3.63	V
V_{PAD}	Voltage at IO	-0.3		V_{DVDD}+0.3	V
V_{IL}	Input Low Voltage	-0.3		0.25*V_{DVDD}	V
V_{IH}	Input High Voltage	0.625*V_{DVDD}		V_{DVDD}+0.3	V
V_{T+}	Schmitt trig low to high threshold point	0.9		2.1	V
V_{T-}	Schmitt trig high to low threshold point	0.7		1.9	V
R_{PU}	Pull-up Resistor	23	29	35	kΩ
R_{PD}	Pull-down Resistor	21	26	31	kΩ
V_{OL}	Output low voltage				V

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V_{OH}	Output high voltage					V
I_{OL}	Low level output current	40pF	12.9	18.2	23.6	mA
		25pF	10.1	14.2	18.4	mA
		12pF	8.4	11.9	15.4	mA
		5pF	5.6	7.9	10.2	mA
I_{OH}	High level output current	40pF	12.2	16.3	20.8	mA
		25pF	9.5	12.7	16.2	mA
		12pF	8.1	10.9	13.9	mA
		5pF	5.4	7.2	9.3	mA

Table 3-9 DC characteristics for VDDIO_SD supplied pins for 1.8V application

Symbol	Parameter		Min	Typical	Max	Unit
V_{DVDD}	I/O supply voltage		1.7	1.8	1.95	V
V_{PAD}	Voltage at IO		-0.3		V_{DVDD}+0.3	V
V_{IL}	Input Low Voltage		-0.3		0.58	V
V_{IH}	Input High Voltage		1.27		2.00	V
V_{T+}	Schmitt trig low to high threshold point		0.4*V_{DVDD}		0.7*V_{DVDD}	V
V_{T-}	Schmitt trig high to low threshold point		0.3*V_{DVDD}		0.6*V_{DVDD}	V
R_{PU}	Pull-up Resistor		23	29	35	kΩ
R_{PD}	Pull-down Resistor		22	26	32	kΩ
V_{OL}	Output low voltage					V
V_{OH}	Output high voltage					V
I_{OL}	Low level output current	TypeA	17.3	24.5	31.8	mA
		TypeB	14.5	20.6	26.6	mA
		TypeC	12.8	18.2	23.6	mA
		TypeD	10.0	14.2	18.4	mA
I_{OH}	High level output current	TypeA	16.2	21.7	27.7	mA
		TypeB	13.5	18.1	23.1	mA
		TypeC	12.2	16.3	20.8	mA
		TypeD	9.5	12.7	16.2	mA

Table 3-10 DC characteristics for VDDIO18 supplied pins for 1.8V application

Symbol	Parameter	Min	Typical	Max	Unit
VDDIO	I/O supply voltage	1.62	1.8	1.98	V
V_{IL}	Input Low Voltage	-0.3		0.35* VDDIO	V
V_{IH}	Input High Voltage	0.65* VDDIO		VDDIO +0.3	V
V_T	Threshold point	0.83	0.91	1	V
V_{T+}	Schmitt trig low to high threshold point	0.95	1.03	1.12	V
V_{T-}	Schmitt trig high to low threshold point	0.73	0.8	0.9	V
V_{TPU}	Threshold point with pull-up resistor enabled	0.82	0.9	1	V
V_{TPD}	Threshold point with pull-down resistor enabled	0.84	0.92	1	V
V_{TPU+}	Schmitt trig low to high threshold point with pull-up resistor enabled	0.94	1.02	1.11	V
V_{TPU-}	Schmitt trig high to low threshold point with pull-down resistor enabled	0.72	0.79	0.89	V

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V_{TPD+}	Schmitt trig low to high threshold point with pull-down resistor enabled		0.96	1.05	1.12	V
V_{TPD-}	Schmitt trig high to low threshold point with pull-up resistor enabled		0.74	0.81	0.91	V
I_L	Input Leakage Current @ V_I=1.8V or 0V				±10	μA
I_{OZ}	Tri-State output leakage current @ V_I=1.8V or 0V				±10	μA
R_{PU}	Pull-up Resistor		55	79	122	kΩ
R_{PD}	Pull-down Resistor		51	87	169	kΩ
V_{OL}	Output low voltage				0.45	V
V_{OH}	Output high voltage		VDDIO -0.45			V
I_{OL}	Low level output current	(DS1,DS0) = 00	7.7	12.8	17.8	mA
		(DS1,DS0) = 01	15.3	25.4	35.2	mA
		(DS1,DS0) = 10	22.8	37.6	51.7	mA
		(DS1,DS0) = 11	30.2	49.2	67.1	mA
I_{OH}	High level output current	(DS1,DS0) = 00	4.8	10.9	18.9	mA
		(DS1,DS0) = 01	9.6	21.5	37.4	mA
		(DS1,DS0) = 10	14.4	32.3	56.0	mA
		(DS1,DS0) = 11	19.1	42.8	74.2	mA

3.4 Audio Codec Electrical Characteristics

Table 3-11 Audio CodecElectrical Specification

Parameter	Symbol	Condition	Min	Typ	Max	Unit
Microphone Bias						
Bias Voltage	V_{MICB}		0.8*AVDD		0.975*AVDD	V
Bias Current	I_{MICB}				3	mA
Microphone Gain Boost PGA						
Programmable Gain	G_{BST}		0		20	dB
Input Resistance	R_{IN}		8		44	KΩ
Input Capacitance	C_{IN}			10		pF
ALC PGA						
Programmable Gain	G_{ALC}		-18		28.5	dB
Gain Step Size				1.5		dB
ADC						
Signal to Noise Ratio	SNR	A-weighted		90		dB
Total Harmonic Distortion	THD	-3dBFS input		-80		dB
Channel Separation				90		dB
Headphone Output Driver						
Programmable Gain	G_{DRV}		-39		6	dB
Gain Step Size				1.5		dB
Output Resistance	R_{OUT}				1	Ω
Output Capacitance	C_{OUT}			20		pF

3.5 SARADC Electrical Characteristics

Table 3-12 SARADC Electrical Specification

Parameter	Symbol	Condition	Min	Typ	Max	Unit
Operating Condition						
Analog Supply	AVDD	-	1.62	1.8	1.98	V

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SARADC Performance						
Resolution	-	-	-	12	-	bit
Effective Number of Bit	ENOB	-	10-	11	-	bit
Differential Nonlinearity	DNL	-	-1.5	-	+1.5	LSB
Integral Nonlinearity	INL	-	-3	-	+3	LSB
Reference voltage	VREFP		-	1.8	-	V
Input Capacitance	C_{IN}	-	-	16	-	pF
Sampling Rate	f_S	-	-	-	2	MS/s
Spurious Free Dynamic Range	SFDR	f_S=2MS/s f_{OUT}=1.17KHz	-	70	-	dB
Signal to Noise and Harmonic Ratio	SNDR	-	-	68	-	dB
Power Consumption						
Analog Supply Current	IAVDD	f_S=2MS/s	-	1.9	-	mA
		Power Down	-	1	-	uA
Digital Supply Current	IVDD	f_S=2MS/s	-	0.1	-	mA
		Power Down	-	1	-	uA

3.6 Power On, Reset and BOOT

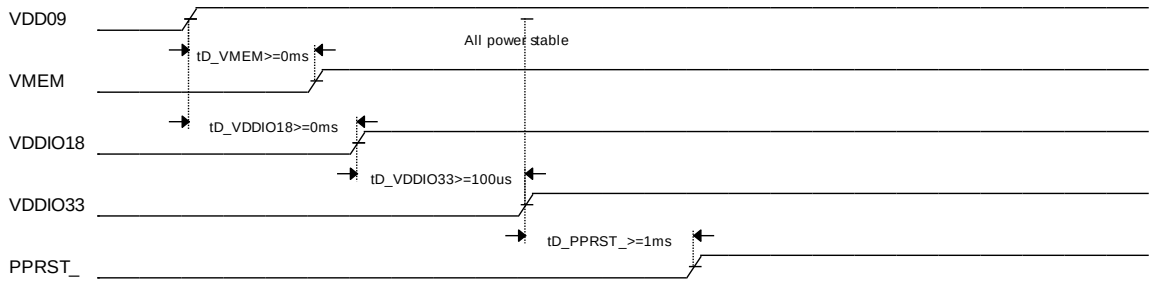
3.6.1 Power-On Timing

The external voltage regulator and other power-on devices must provide the X2600 processor with a specific sequence of power and resets to ensure proper operation. Following are the name of the power：

- VDD09: all 0.9V power supplies
- VMEM: VDDMEM
- VDDIO18: all 1.8V power supplies, VDDIO18, PLL_AVDD, SARADC_AVDD, DDR_PLLVCCA, CODEC_AVDD, TX_VCCA18, USB_AVD18, EFUSE_AVDD, VDDIO_CIM[Note1], VDDIO_SD[Note2]
- VDDIO33: all 3.3V power supplies, VDDIO33, USB_AVD33, VDDIO_CIM[Note1], VDDIO_SD[Note2]

Table 3-13 Power-On Timing Parameters

Symbol	Parameter	Min	Max	Unit
t_{D_VMEM}	Delay between VDD09 arriving 90% to VMEM arriving 50%	0	–	ms
$t_{D_VDDIO18}$	Delay between VDD09 arriving 90% to VDD18 arriving 50%	0	–	ms
$t_{D_AVDIO33}$	Delay between VDDIO18 arriving 90% to VDDIO33 to be turned on	100	–	us
$t_{D_PPRST_}$	Delay between all power rails get stable and power-on reset PPRST_ de-asserted	1	–	ms[2]



Note1: VDDIO_CIM supports two level standards, 3.3V and 1.8V.

Note2: VDDIO_SD supports two level standards, 3.3V and 1.8V.

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Figure 3-1 Power-On Timing Diagram

3.6.2 Power On Reset

X2600 has an internal POR circuit, and compares PLL_AVDD,VDD with the reference voltage to generate a signal which is open-drain output.

Table 3-14 POR

Symbol	Parameter	Min	Typical	Max	Unit
V_{tha+}	Power up threshold on PLL_AVDD supply[1]	1.3	1.35	1.4	V
V_{tha-}	Power down threshold on PLL_AVDD supply[1]	1.2	1.25	1.3	V
V_{thd+}	Power up threshold on VDD supply[1]	0.55	0.6	0.65	V
V_{thd-}	Power down threshold on VDD supply[1]	0.45	0.5	0.55	V
V_{hysa}	POR hysteresis on PLL_AVDD supply[1]	-	0.1	-	V
V_{hysd}	POR hysteresis on VDD supply[1]	-	0.1	-	V
T_{dr}	Debounce delay time for POR rising edge	-	120	-	us
T_{df}	Debounce delay time for POR falling edge	-	4	-	us
T_{D_POR}	Extend delay time for POR rising edge[2]		15		ms

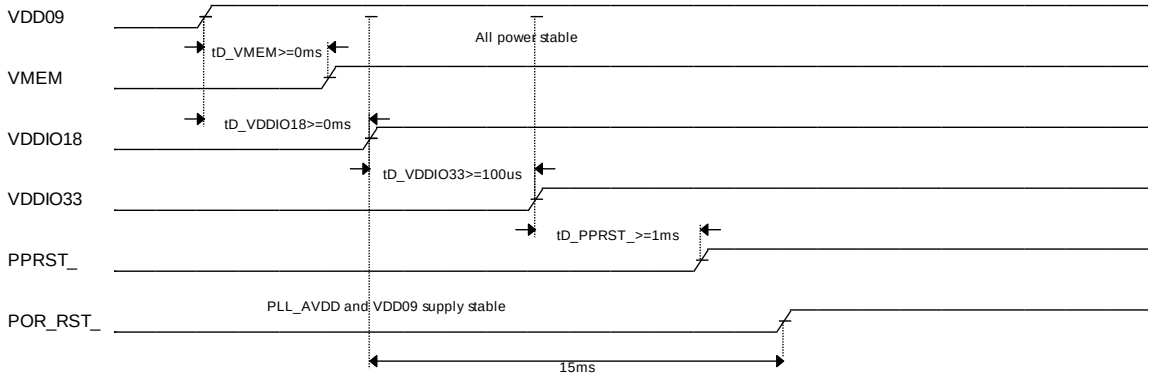


Figure 3-2 POR_RST_ Timing Diagram

Note:

1 In order to use the POR circuit to reset X2600, follow the circuit design below to connect the POR_RST_ to PPRST_.

2 After PLL_AVDD and VDD supply stable, the POR_RST_ output will be extended 15ms then release..

Following figure shows the sequence where PLL_AVDD rises first and VDD drops first.

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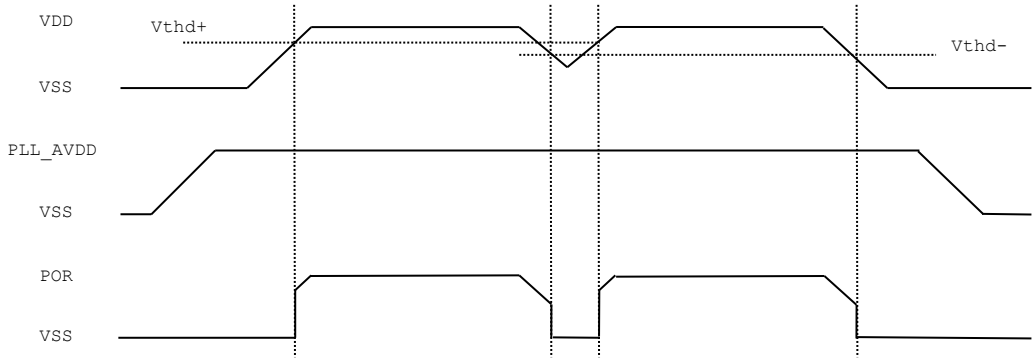


Figure 3-3 POR TIMING 1

Following figure shows the sequence where VDD rises first and PLL_AVDD drops first.

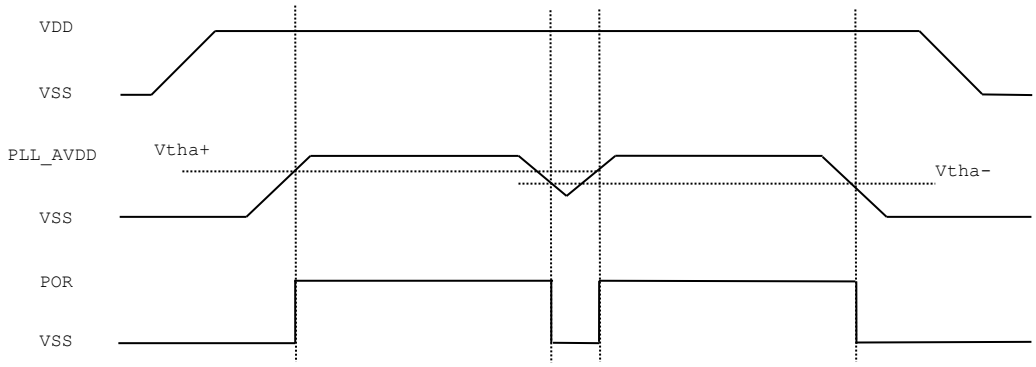


Figure 3-4 POR TIMING 2

3.6.3 Reset procedure

In addition, X2600 also supports 2 reset sources: 1 PPRST_ pin reset; 2 WDT timeout reset. After reset, program start from boot.

1. PPRST_ pin reset

This reset is trigged when PPRST_ pin is put to logic 0. It happens in power on all power stable and RESET-KEY pressed to reset the chip from unknown dead state.

2. WDT timeout reset

This reset happens in case of WDT timeout. The reset keeps for about a few EXCLK/512 cycles. After reset, all GPIO shared pins are put to GPIO input function, except JTAG relate TCK/ TMS/ TDI/TDO which reset to function mode. The oscillators are on

3.6.4 BOOT

The boot sequence of the X2600 is controlled by boot_sel [2:0].

Table 3-15 Boot configuration of X2600

boot_sel[2]	boot_sel[1]	boot_sel[0]	Boot configuration	Error jump
0	0	0	MSC0(PD0)_3.3V	SFC(PD0)_3.3V
0	0	1	SFC(PD6)_3.3V	MSC1(PD6)_3.3V

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0	1	0	SFC(PD0)_3.3V	MSC1(PC25)_3.3V
0	1	1	USB	
1	0	1	MSC0(PD0)_1.8V	SFC(PD0)_1.8V
1	1	0	SFC(PD0)_1.8V	USB
1	1	1	USB	

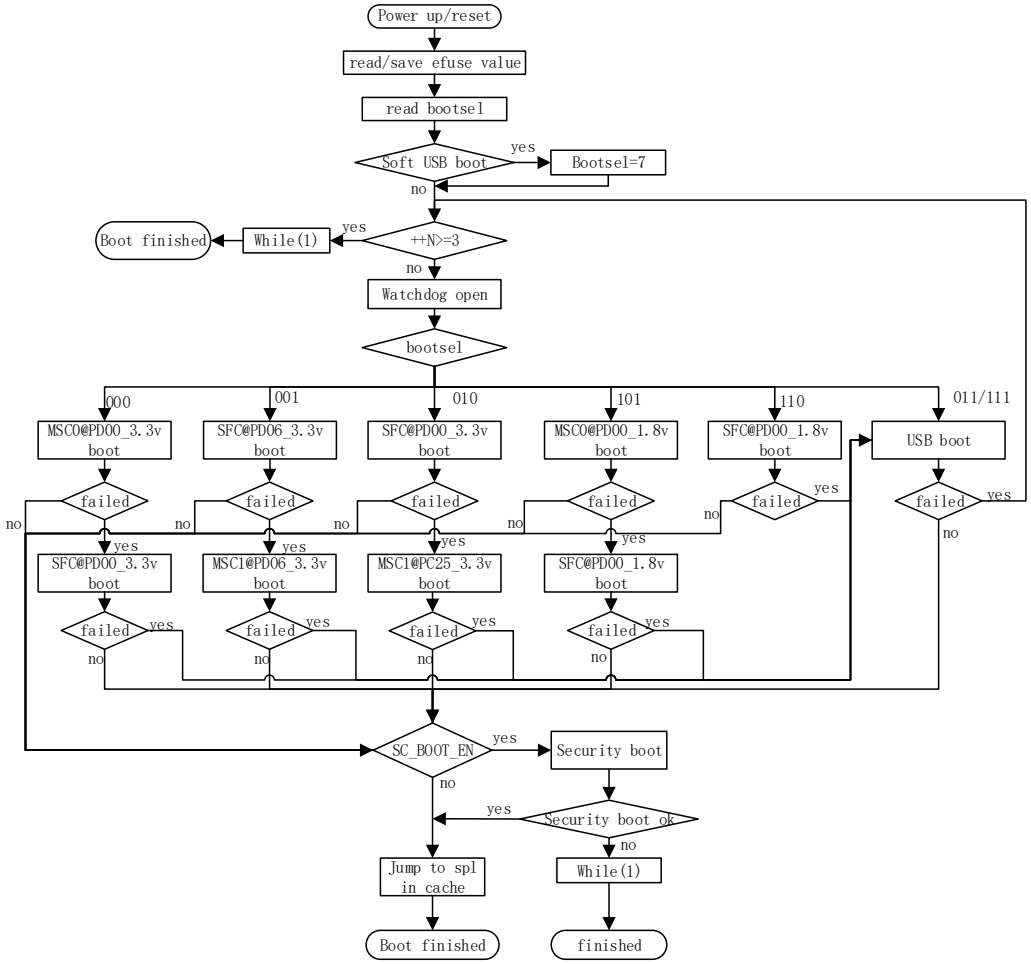


Figure 3-5 Boot sequence diagram of X2600

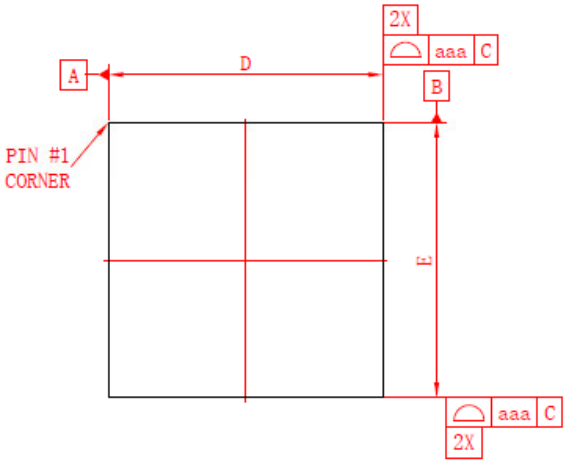
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4 Packaging Information

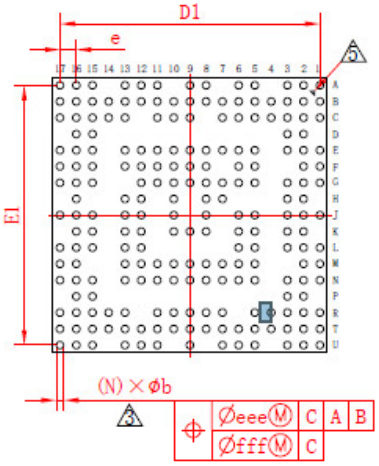
4.1 Overview

X2600 processor is offered in 209-pin BGA package, which is 11mm * 11mm * 1.18mm outline, 17 * 17 matrix ball grid array and 0.65mm ball pitch, show in **Figure 4-1**.

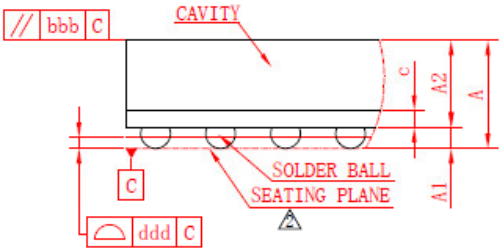
4.2 Device Dimensions



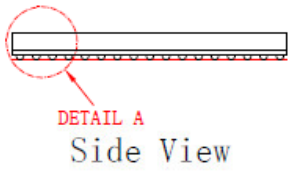
Top View



Bottom View



DETAIL A



Side View

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symbol	Dimension in mm			Dimension in inch		
	MIN	NOM	MAX	MIN	NOM	MAX
A	---	---	1.180	---	---	0.046
A1	0.160	0.210	0.260	0.006	0.008	0.010
A2	0.820	0.870	0.920	0.032	0.034	0.036
c	0.140	0.170	0.200	0.006	0.007	0.008
D	10.900	11.000	11.100	0.429	0.433	0.437
E	10.900	11.000	11.100	0.429	0.433	0.437
D1	---	10.400	---	---	0.409	---
E1	---	10.400	---	---	0.409	---
e	---	0.650	---	---	0.026	---
b	0.250	0.300	0.350	0.010	0.012	0.014
aaa	0.100			0.004		
bbb	0.100			0.004		
ddd	0.080			0.003		
eee	0.150			0.006		
fff	0.080			0.003		
Ball Diam	0.300			0.012		
N	209			209		
MD/ME	17/17			17/17		

Figure 4-1 x2600 package outline drawing

Notes:

1. BALL PAD OPENING: 0.270mm;
2. PRIMARY DATUM C AND SEATING PLANE ARE THE SOLDER BALLS;
3. DIMENSION b IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER,PARALLEL
4. TO PRIMARY DATUM C;
5. SPECIAL CHARACTERISTICS C CLASS: bbb,ddd;
6. THE PATTERN OF PIN 1 FIDUCIAL IS FOR REFERENCE ONLY;
7. BAN TO USE THE LEVEL 1 ENVIRONMENT-RELATED SUBSTANCES OF JCET PRESCRIBING;

4.3 Solder Ball Materials

Both the top (joint) and bottom solder ball materials of X2600 are SAC125.

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4.4 Moisture Sensitivity Level

X2600 package moisture sensitivity is level 3.

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